

PROBLEM STATEMENTS - SYSTEM SEIGE

DOMAINS:

1. Finance
2. Edu Tech
3. Health Tech
4. Cyber Security & Web Mining
5. AI / ML & Blockchain
6. Marketing, Analytics & Social

Any component of your solution that invokes an LLM or generative AI model (chatbots, agents, scoring, classification, recommendation, or risk-analysis features) must use your own API key (BYOK), with the provider and model version disclosed in your repo README. Hardcoded, mocked, or rule-based logic disguised as AI functionality is not permitted and will be flagged as a build-phase violation. This applies regardless of whether AI is core to your PS or an added feature

Finance:

PS - 001. Autonomous Multi-Region Payment & Identity Settlement Network

Problem Statement: Develop a sophisticated, multi-region distributed network for atomic payment settlement. The platform must implement real-time, privacy-preserving fraud detection by synthesizing federated identity signals across regions, offer interpretable/explainable fraud risk scoring, and maintain independently verifiable cryptographic reconciliation proofs to ensure compliance and auditability across all operational jurisdictions.

PS - 002 .AI-Powered Financial Wellness & Loan Transparency Platform

Problem Statement: Build an AI-powered financial intelligence platform that empowers individuals to make informed financial decisions by combining personalized financial guidance with transparent loan analysis. The platform should leverage a multi-agent financial advisory system, where specialized AI agents collaboratively assist users in budgeting, debt management, savings, investments, insurance, taxation, and loan evaluation to deliver holistic and context-aware financial recommendations. It should help users understand their financial health, identify hidden risks in financial products, analyze loan agreements for unfair charges and regulatory compliance issues, and generate actionable recommendations to improve long-term financial stability. The platform should support multilingual interactions, simplify complex financial documents, and provide explainable insights that promote financial literacy, consumer protection, and responsible financial decision-making while helping users avoid debt traps and predatory lending practices.

Edu Tech:

PS - 003. AI-Powered Secure Assessment & Integrity Platform

Problem Statement: Build a next-generation AI-powered online assessment platform capable of conducting large-scale coding assessments, aptitude tests, technical MCQs, and certification exams while ensuring exam integrity using multimodal AI proctoring, behavioral analytics, and explainable risk scoring. The platform should provide end-to-end assessment lifecycle management from question creation and secure delivery to AI-powered evaluation, cheating detection, analytics, and candidate skill profiling. Unlike traditional online judges, the system must intelligently detect suspicious behavior, identify plagiarism, generate detailed candidate insights, and help recruiters understand not only what score a candidate achieved, but how they arrived at that score.

Health Tech:

004. Multi-Agent Negotiation for Emergency Resource Reallocation

Problem Statement: Build an intelligent multi-agent decision-making platform that enables hospitals to rapidly and transparently reallocate critical resources during emergencies and operational disruptions. The system should represent hospital resources including operating rooms, medical staff, ICU beds, equipment, and emergency facilities as autonomous agents capable of negotiating resource allocation using explainable bidding or scoring mechanisms. It should model interdependencies between resources to evaluate the downstream impact of every reallocation decision, dynamically prioritize critical cases, and generate fast, fair, and explainable allocation strategies. The platform should provide real-time visualization of the negotiation process, enabling healthcare administrators to understand why specific decisions were made, optimize resource utilization, minimize treatment delays, and improve overall emergency response efficiency.

Cyber Security & Web Mining:

005. Website Defacement Detection & Vulnerability Assessment Platform

Build an AI-powered web security and monitoring platform that enables organizations to continuously protect, monitor, and assess the security posture of their web applications and digital assets. The platform should detect website defacement, security vulnerabilities, and suspicious behavioral changes in real time through continuous monitoring and intelligent analysis. It should provide instant alerting, visual change detection between snapshots, robust role-based access control, and comprehensive audit trails while supporting centralized management of multiple protected assets. Additionally, the platform should leverage AI-driven security intelligence to prioritize risks, identify emerging threats, recommend remediation actions, and help security teams proactively respond to incidents, strengthen compliance, and improve overall cyber resilience.

006. Enterprise Multi-Tenant Secrets & Certificate Lifecycle Platform

Problem Statement: Build a highly secure, enterprise-grade platform for centralized secrets and certificate lifecycle management that enables organizations to securely store, manage, and automate the rotation of cryptographic assets across cloud-native and on-premises environments. The platform should maintain an immutable, tamper-evident audit trail for every access and modification, enforce complete tenant isolation through fine-grained access controls, and provide automated certificate renewal and expiration monitoring. Additionally, it should leverage AI-powered secret exposure detection to continuously scan source code repositories, configuration files, logs, CI/CD pipelines, and enterprise documents for leaked credentials, API keys, certificates, tokens, and other sensitive information. The system should intelligently prioritize risks, recommend remediation actions, and help security teams proactively prevent credential compromise while ensuring regulatory compliance and operational resilience.

AI, ML & Blockchain:

007. Smart Tourist Safety Monitoring & Incident Response System using AI, Geo-Fencing, and Blockchain-based Digital ID

Problem Statement: Create an intelligent tourist safety and emergency response platform that leverages Artificial Intelligence, geo-fencing, and blockchain-based digital identities to ensure the safety of tourists in real time. The system should proactively detect emergencies, monitor movement within predefined safe zones, provide instant SOS and incident response, and enable secure identity verification while preserving user privacy. It must offer multilingual AI assistance, predictive risk analysis, and a centralized dashboard for authorities to monitor incidents, coordinate emergency services, and improve overall tourist safety across cities and destinations.

008. Automated warranty/insurance claim damage assessor

Problem Statement: Develop an AI driven damage assessment platform that automates the vehicle insurance and warranty claim process by analyzing images of damaged vehicles. The system should use computer vision to detect and classify damage, estimate repair costs based on standardized pricing data, and generate an explainable claim recommendation with a confidence score. Additionally, it should identify potentially fraudulent claims through duplicate image detection, metadata analysis, and anomaly detection, enabling insurers to accelerate claim processing, reduce manual inspections, and improve decision accuracy.

Marketing, Analytics & Social:

009. Multi-Vendor Marketplace & Logistics Management Platform

Problem Statement: Build an AI-powered enterprise marketplace platform that enables seamless collaboration between buyers, independent sellers, warehouses, and logistics providers within a unified digital ecosystem. The platform should support end-to-end order lifecycle management, maintain real-time inventory consistency across distributed vendors, enforce hierarchical role-based access control, and maintain immutable audit trails for financial, operational, and inventory activities. It should leverage intelligent decision-making to optimize inventory planning, demand forecasting, order fulfillment, logistics coordination, and vendor performance, enabling organizations to improve operational efficiency, supply chain resilience, and customer experience at scale.

010. Enterprise Collaboration & Hierarchical Access Management Platform

Problem Statement: Build an AI-powered enterprise collaboration platform that enables organizations to securely manage documents, projects, workflows, and team collaboration within a unified digital workspace. The platform should support dynamic hierarchical access control with fine-grained authorization across departments, teams, projects, and individual resources while maintaining immutable, tamper-evident audit trails for all user activities. It should incorporate intelligent access governance to continuously analyze permission structures, identify excessive or anomalous access, recommend least-privilege policies, and strengthen organizational security without disrupting collaboration. By combining secure information sharing, adaptive access management, and enterprise-grade governance, the platform should improve operational efficiency, regulatory compliance, and data protection across large-scale organizations.